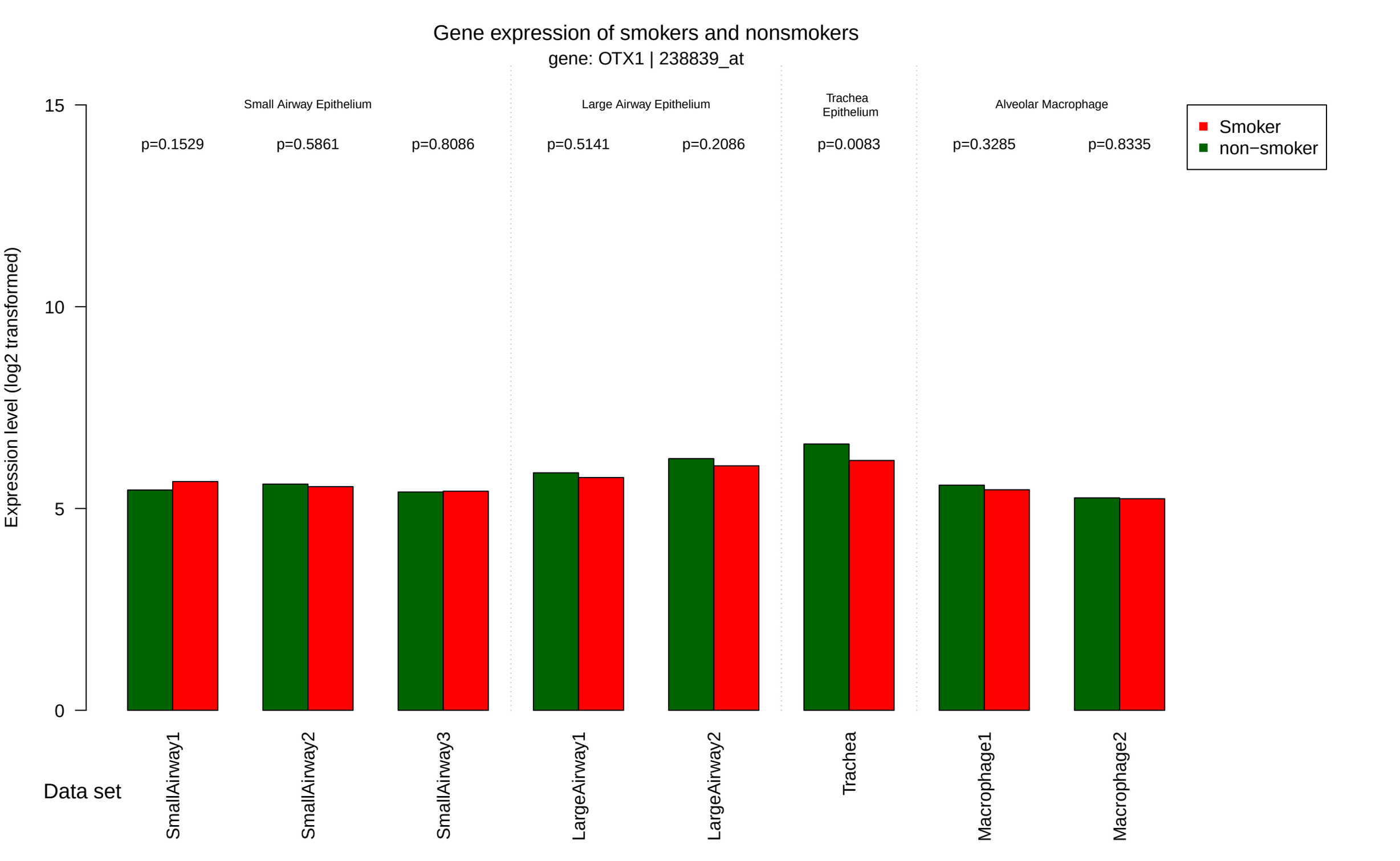
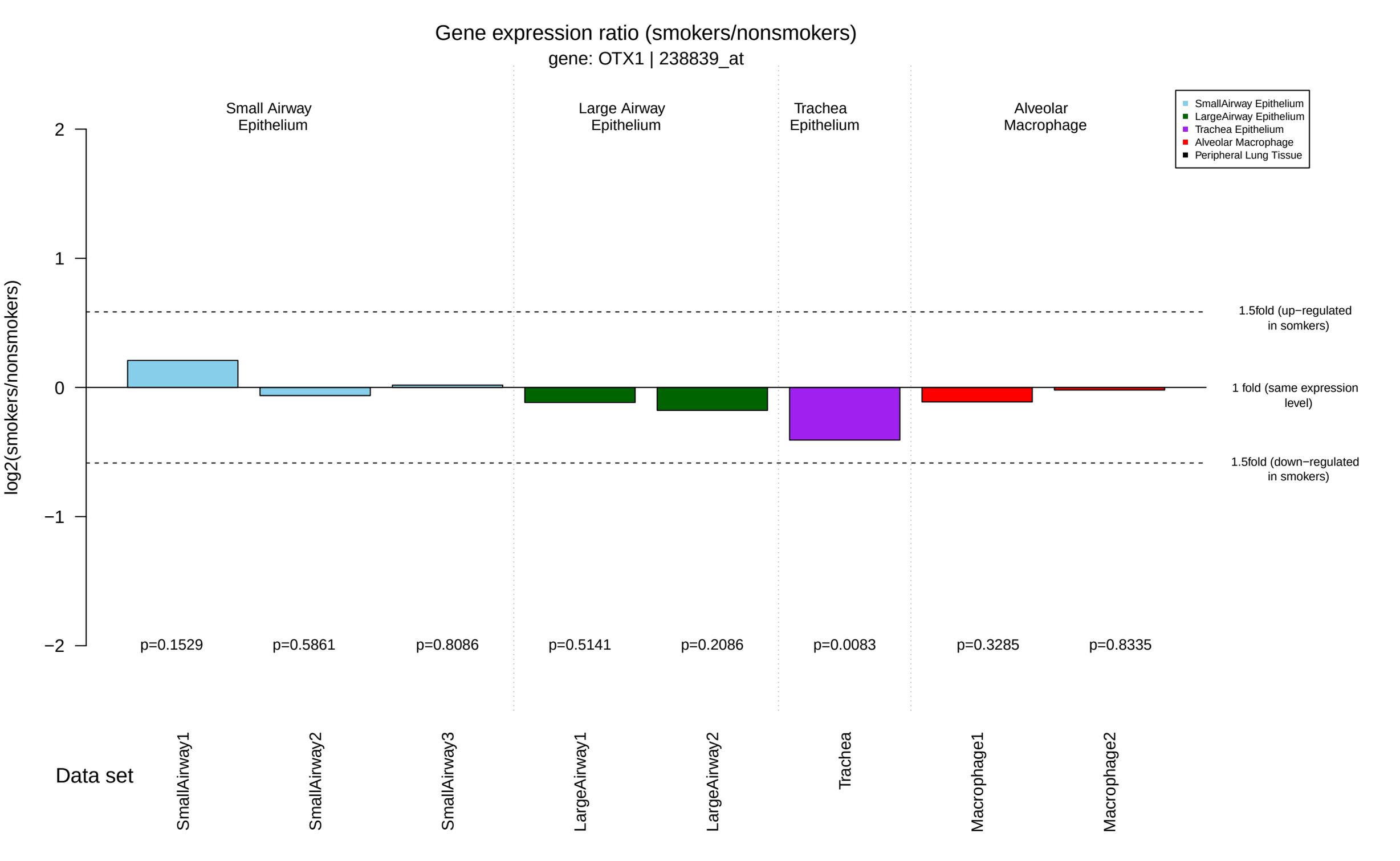
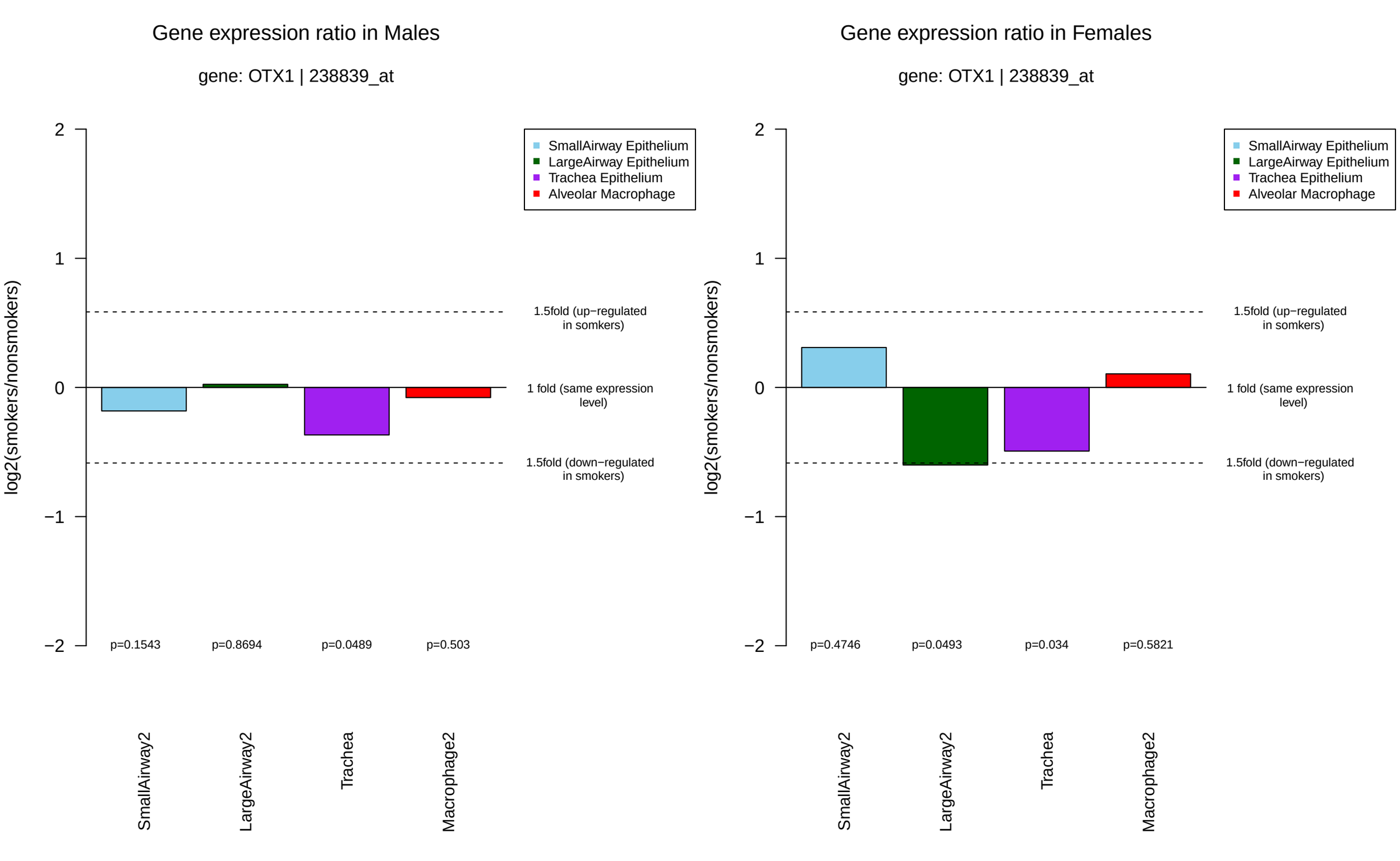
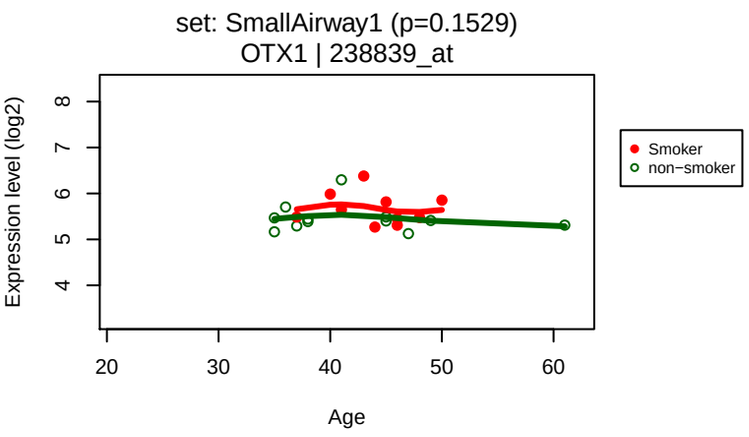
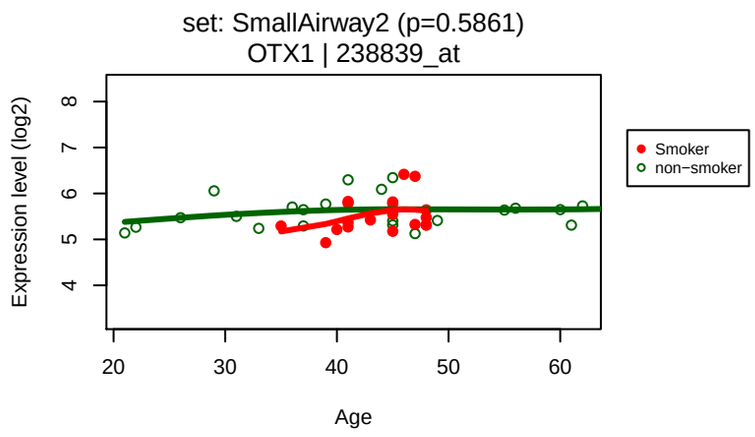
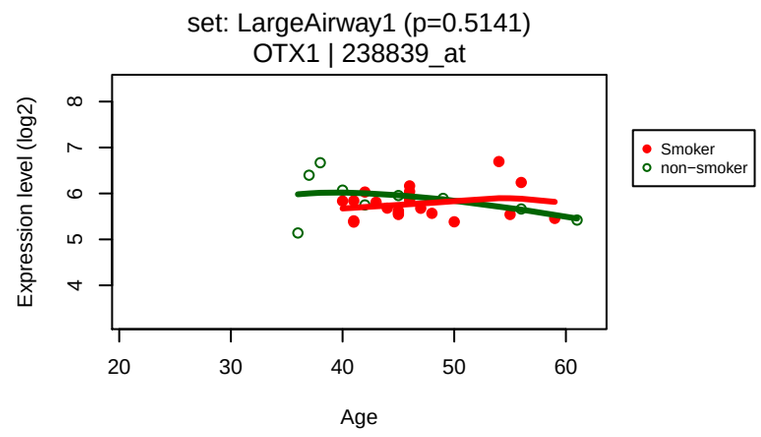

Gene expression of smokers and nonsmokers
gene: OTX1 | 238839_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1529
p=0.5861
p=0.8086
p=0.5141
p=0.2086
p=0.0083
p=0.3285
p=0.8335
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

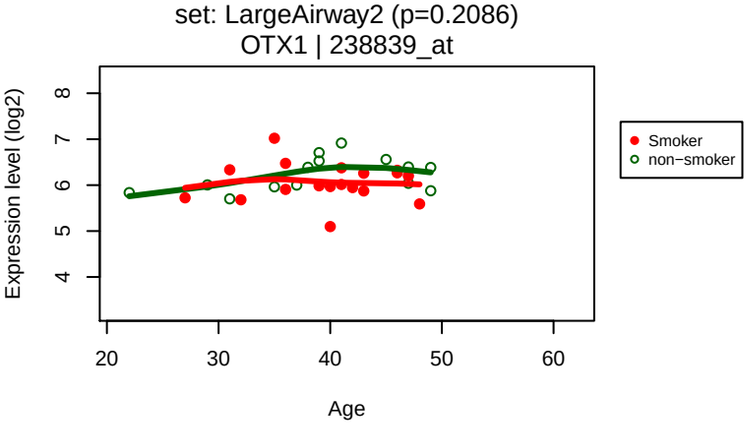

Gene expression ratio (smokers/nonsmokers)
gene: OTX1 | 238839_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1529
p=0.5861
p=0.8086
p=0.5141
p=0.2086
p=0.0083
p=0.3285
p=0.8335
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

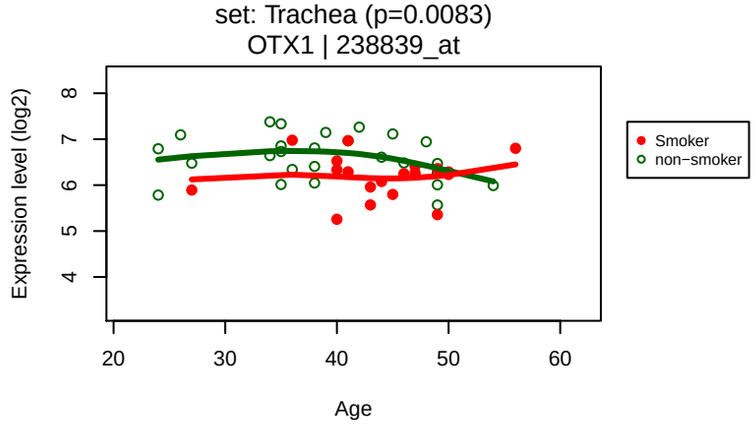

Gene expression ratio in Males
gene: OTX1 | 238839_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1543
p=0.8694
p=0.0489
p=0.503
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OTX1 | 238839_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4746
p=0.0493
p=0.034
p=0.5821
SmallAirway2
LargeAirway2
Trachea
Macrophage2

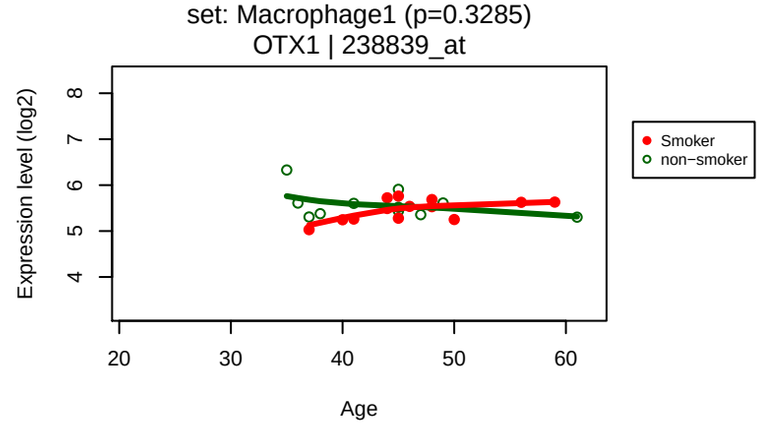

set: SmallAirway1 (p=0.1529)
OTX1 | 238839_at
Expression level (log2)
Age
Smoker
non−smoker

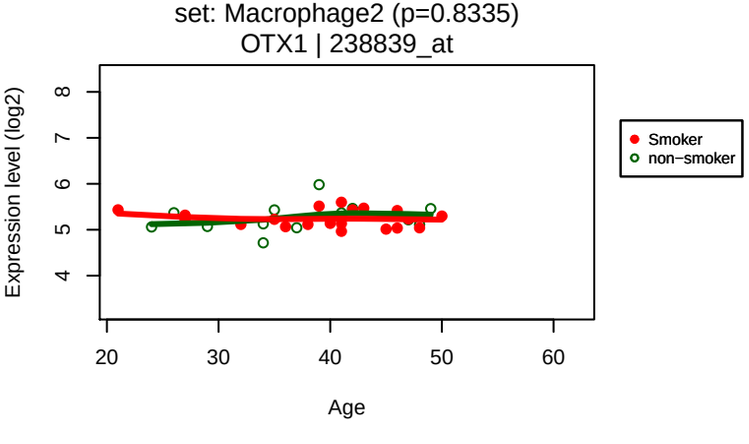

set: SmallAirway2 (p=0.5861)
OTX1 | 238839_at
Expression level (log2)
Age
Smoker
non−smoker

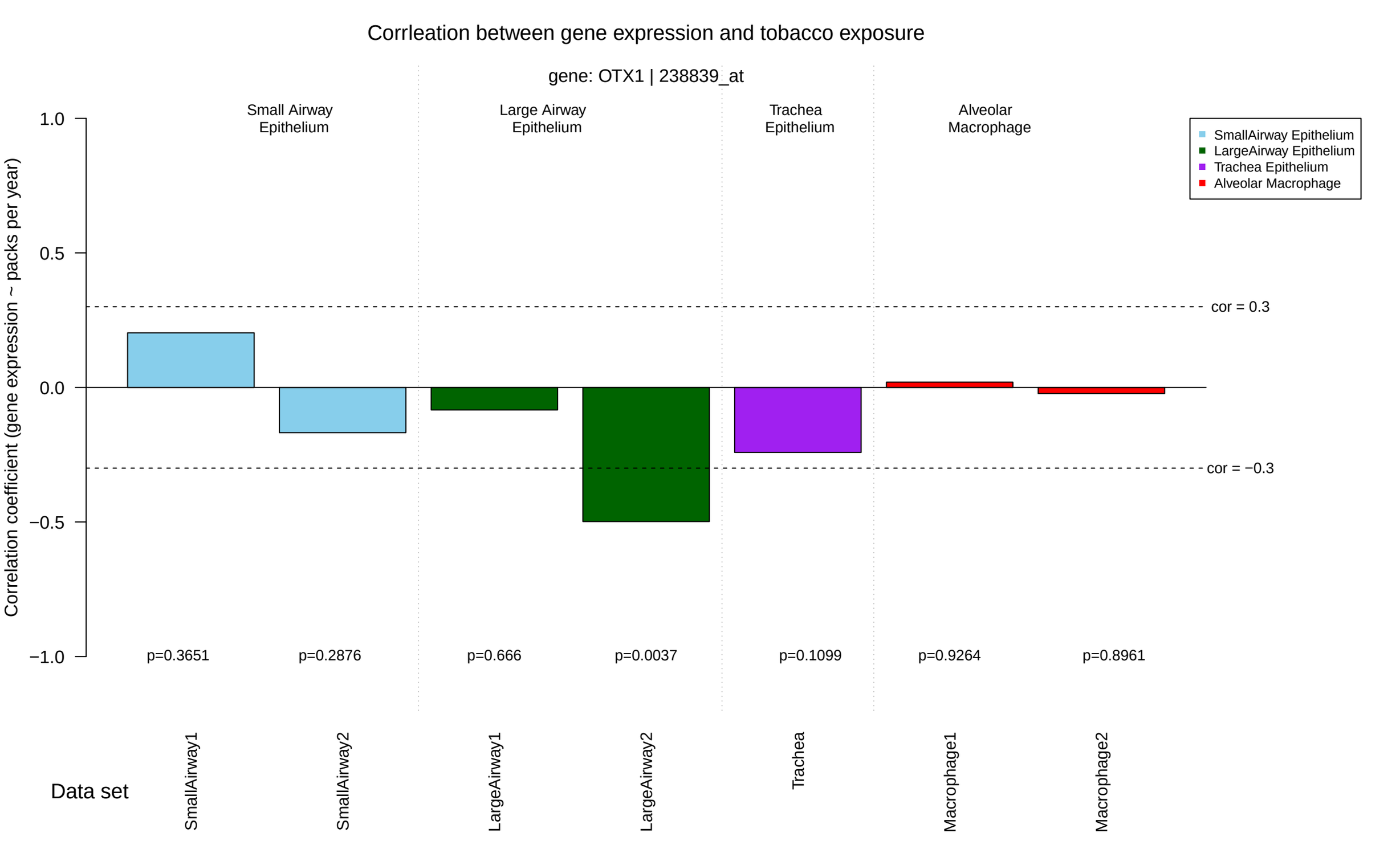

set: LargeAirway1 (p=0.5141)
OTX1 | 238839_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2086)
OTX1 | 238839_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0083)
OTX1 | 238839_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3285)
OTX1 | 238839_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8335)
OTX1 | 238839_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OTX1 | 238839_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3651
p=0.2876
p=0.666
p=0.0037
p=0.1099
p=0.9264
p=0.8961
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2